

A Reconstruction Methodology for Marginal Cost and Marginal CO₂ Emissions in the Chilean Power System Using CEN Public Data

Marcelo Matus A., *Member, IEEE*, María del Pilar Buitrago Villada, Erick Sierra Baeza, Carlos Enrique García Bujanda, Marcos Falcone Salazar, and Gonzalo Bustos Turu

Abstract—The decarbonisation of the Chilean power sector and the country’s commitment to a carbon-tax escalation under Law 20.780 require operational signals that are both *spatially* resolved (per-bus) and *marginally* attributed (per-unit) — the natural analogue of a Locational Marginal Emission factor. At present the Coordinador Eléctrico Nacional (CEN) publishes only a 15-minute nodal marginal cost λ^{CEN} and an annual system average emission factor; no per-bus, per-quarter marginal CO₂ intensity ε is released. This paper presents *cen2gtopt*, a novel reconstruction methodology that re-engineers ε for the 29 CEN-published 220/500 kV buses at 15-minute resolution from public data alone. The procedure combines the SCVIC merit-ladder of declared variable costs, the “generación real” hourly dispatch, the “instrucciones operacionales” price-taker classification, and the Marginal Loss Factors (MLFs) extracted from the PLEXOS PCP/PID solution .acddb oracle. A chained-tolerance plus cumulative-spread clustering rule separates the 29 buses into electrically coherent islands per quarter; for each island the marginal unit is identified as the dispatched non-forced generator whose declared CV is closest to the median bus marginal price. The methodology is applied to the operating day 2026-04-22. Across 2494 (bus, quarter) cells, the reconstructed marginal cost $\hat{\lambda}$ agrees with λ^{CEN} to a mean absolute error of 0.152 USD MWh⁻¹, a 90th percentile of 0.471 USD MWh⁻¹ and a maximum of 1.338 USD MWh⁻¹; after excluding

the 16.0% of cells flagged as *tier_uncertainty*, the mean drops to 0.124 USD MWh⁻¹ and the maximum to 1.338 USD MWh⁻¹. Per-island marginal-unit coherence is exact (0 of 474 islands disagree). The pipeline runs in approximately 6 s on a warm cache and is published as part of the *gtopt* project.

Index Terms—Locational marginal emissions, marginal CO₂ factor, locational marginal price, Coordinador Eléctrico Nacional, SCVIC, marginal loss factor, merit ladder, public-data reconstruction, short-run-marginal emissions, Chilean power system.

I. INTRODUCTION

The decarbonisation of electricity grids has elevated the marginal emission factor (MEF) from a research curiosity into an operational signal used for demand-response orchestration, electric-vehicle charging optimisation, hydrogen-production scheduling, and corporate scope-2 carbon accounting [?], [?], [?]. In hubs that publish nodal marginal prices, the MEF is increasingly being constructed as a direct analogue of the Locational Marginal Price (LMP): per pricing node, per dispatch interval, with the same energy/loss/congestion decomposition that Schweppe *et al.* [?] formalised for spot prices and Hogan [?] extended to nodal pricing. PJM Interconnection now publishes five-minute Locational Marginal Emissions (LME) for CO₂, NO_x, and SO₂ across approximately 1200 pricing nodes [?]; CAISO offers an emissions tab on its operator dashboard [?]; WattTime broadcasts a per-balancing-area Marginal Operating Emission Rate with a 72-hour forecast [?].

a) Chilean context: The Chilean Coordinador Eléctrico Nacional (CEN) clears the Sistema Eléctrico Nacional (SEN) every 15 minutes via a PLEXOS-based day-ahead Programa de Coordinación de la Producción (PCP) plus intra-day Programa Intra-Diario (PID). CEN publishes the resulting nodal marginal cost λ^{CEN} for over 1300 substation buses through its public Sistema de Información Pública (SIP) API [?]. Generators submit the Sistema de Costos Variables Informados de Centrales (SCVIC) — a piecewise-linear declaration of variable

Manuscript received MONTH DAY, 2026; revised MONTH DAY, 2026.

M. Matus A. is with Black Bear Engineering SpA, Santiago, Chile, and the Centro Nacional de Inteligencia Artificial (CENIA, FB210017), Santiago, Chile (e-mail: marcelo.matus@blackbear.cl).

M. P. Buitrago Villada is with the Department of Electrical Energy and Automation, Universidad Nacional de Colombia, Medellín, Colombia.

E. Sierra Baeza is with Black Bear Engineering SpA, Santiago, Chile, and CENIA.

C. E. García Bujanda is with Black Bear Engineering SpA, Santiago, Chile.

M. Falcone Salazar is with the School of Engineering, Universidad Adolfo Ibáñez, Santiago, Chile, and formerly with Colbún S.A., Santiago, Chile.

G. Bustos Turu is with the Department of Chemical Engineering, Imperial College London, London, UK, and the Centre for Energy, Universidad de Chile, Santiago, Chile.

cost by configuration, fuel, and load tier — and the Coordinador publishes the resulting ladders as parquet datasets. Finally, the PLEXOS solution databases (.accdb) for both PCP and PID are publicly archived, exposing per-period Marginal Loss Factors (MLFs), nodal loads, and unit dispatch.

b) The gap: What CEN does *not* publish is a per-bus, per-quarter marginal emission factor ε . The Ministry of Energy’s *Indicadores Ambientales* page only releases an *annual* average emission factor for the SEN (2020-base, updated yearly) [?]. The Comisión Nacional de Energía (CNE) hosts a legacy SIC/SING zonal visualisation that pre-dates the 2017 unification of Chile’s two former sub-systems. The *Reporte Anual Art. 72* of the Coordinador [?] does report annual emissions, but neither at locational nor at marginal granularity. Yet the Chilean carbon tax under Law 20.780 — currently $\tau_{\text{CO}_2} = 5 \text{ USD/tonneCO}_2$ with escalation under discussion [?] — and the proposed European Carbon Border Adjustment Mechanism are both bus-resolved, marginal-unit-resolved instruments. A locationally and temporally disaggregated ε is therefore a regulatory and commercial necessity, not a research nicety.

c) Contribution: This paper presents **cen2gtopt**, a Python pipeline that reconstructs ε at 29 CEN-published transmission buses for any operating day from public data alone. The contributions are:

- 1) A *merit-ladder picker*: $\hat{\lambda}$ at each island is reconstructed by selecting, among the dispatched non-forced generators, the one whose SCVIC-declared CV is closest in absolute distance to the published median λ^{CEN} (Section ??, Algorithm ??).
- 2) A *chained-tolerance with cumulative-spread cap* for island clustering, parameterised by the pair $(\Delta_{\text{tol}}, S_{\text{max}}) = (\$0.5, \$0.8) / \text{MWh}$, which cures the chained-clustering pathology inherent in pure pairwise λ -grouping (Section ??).
- 3) A *per-bus MLF correction* that uses CEN’s exact PLEXOS-solver-output MLF table (extracted from the PCP/PID .accdb) when available, falling back to an empirical $\lambda_b / \lambda_{\text{ref}}$ ratio (Section ??).
- 4) A *cross-validation oracle*: the same PLEXOS .accdb is parsed for the published per-unit dispatch and per-bus marginal cost, allowing an end-to-end check against the LP-cleared values rather than the publication price only (Section ??).
- 5) An open-source release, available as a pip-installable Python package, with a hash-keyed parquet cache that makes per-day re-runs idempotent and a multi-day orchestrator with Hive-partitioned output for back-testing.

The remainder of the paper is organised as follows. Section ?? reviews the state of the art and the methodological choice points specific to a public-data reconstruction. Section ?? develops the **cen2gtopt** pipeline (with its three operating modes **simulated**, **real**, **real-reconstruct**) and the LP-duality justification of the closest-CV picker under the PLEXOS PID overlay. Section ?? reports a pedagogical IEEE 9-bus example, the production application

to the 29-bus CEN system on 2026-04-22, and a sensitivity analysis. Section ?? consolidates the validation numbers against λ^{CEN} and the PLEXOS oracle. Section ?? surveys structural limitations and the carbon-credit implications of a per-bus per-quarter ε signal. Section ?? closes with the forward-looking integration into the **gtopt** SDDP solver.

II. BACKGROUND AND RELATED WORK

A. Locational marginal pricing

The locational marginal price (LMP) construct on which this work depends derives from the spot-pricing framework of Schweppe *et al.* [?] and the contract-network formulation of Hogan [?], who decomposed

$$\lambda_b = \lambda_{\text{ref}} + \text{LMC}_b + \text{LCC}_b \quad (1)$$

into reference energy, marginal-loss component (LMC), and marginal-congestion component (LCC). The present pipeline absorbs LMC_b into a per-bus loss factor and LCC_b into the island partition itself.

B. Marginal versus average emission factors

Hawkes [?] demonstrated that marginal CO_2 factors can differ from system averages by an order of magnitude and that this gap materially affects energy-efficiency build/no-build decisions. Siler-Evans *et al.* [?] regressed historical EPA generator-level data to fit MEFs at the Reliability Council level and showed strong diurnal/seasonal cycles. Callaway *et al.* [?] formalised the distinction between operational and decision-relevant emission factors; the present paper sits firmly in the operational/short-run camp, computing factors against the realised CEN dispatch.

C. ISO production tools

PJM [?], [?] publishes five-minute per-node emission rates with the same energy/loss/congestion structure as the LMP itself. CAISO [?] surfaces a system-level emission curve at five-minute resolution but is not nodal. WattTime [?] ships a balancing-area Marginal Operating Emission Rate with a 72-hour forecast but is neither nodal nor unit-attributed. NREL’s Cambium [?] publishes *long-run* factors for capacity-expansion studies, not operational dispatch. Electricity Maps [?], [?] publishes hourly per-zone consumption-based factors and discontinued its marginal feed in 2024.

D. Counterfactual marginals

A research-grade refinement is the counterfactual evaluation of Valenzuela *et al.* [?], who exploit implicit differentiation of the QP/LP optimality conditions to recover an exact dynamic locational marginal emission factor from a solved dispatch. The present pipeline is empirical rather than counterfactual, but its open-source structure with the PLEXOS .accdb oracle is designed so that the same input set can be fed into the **gtopt** SDDP solver to produce a counterfactual companion in future work.

E. Prior CEN-specific work

For Chile, Bonvallet *et al.* [?] co-optimised hydrothermal scheduling under non-convex irrigation constraints; Moreno *et al.* [?] treated distributed-generation deployment under uncertainty. The Coordinador's annual *Reporte Art. 72* [?] aggregates emissions at the system level, and the 4eChile guide [?] documents a manual marginal-emission estimation tool for CDM projects. None publishes a per-bus per-quarter signal.

F. Methodological choice points

Three choices dominate the design space. The *merit-stack picker*: should the marginal unit be the last-dispatched cleared at λ_{ref} ("maximum CV $\leq \lambda$ " [?]), or the absolute-closest CV? The pipeline picks the absolute-closest CV because the closest-above is sometimes nearer than the closest-below at low-CMG hours. The *forced-unit treatment*: units flagged with $\text{consigna} \in \{\text{CI, MT, PMT, PP, PS, PC}\}$ are price-takers when at technical minimum but remain valid candidates when interior. The *loss-factor decomposition*: the pipeline uses CEN's published MLF (extracted from the PID .accdb) when available, falling back to the empirical $\lambda_b/\lambda_{\text{ref}}$ ratio when it is not.

III. METHODOLOGY

The pipeline is a directed acyclic graph of stages whose final two nodes are the **cen2gtopt** canonical-feed builder (**gtopt_canonical_feed**) and the marginal-unit identifier (**gtopt_marginal_units**); every CEN/SCVIC/PLEXOS extractor is wrapped by a hash-keyed parquet cache so that any upstream change short-circuits at the first byte that has changed (Fig. ??). Three operating modes are exposed: **simulated** (re-solve a reduced LP), **real** (use the published cleared dispatch), and **real-reconstruct** (the default — the closest-CV picker over the SCVIC merit ladder); a **compare** mode runs all three side-by-side for cross-checking.

A. Notation

Time is partitioned into operating days, hours, and quarters (15 min blocks). $\mathcal{T} = \{0, \dots, 95\}$ indexes quarters; the network exposes a bus set $\mathcal{N}$ at which CEN publishes a nodal marginal cost $\lambda_{b,t}^{\text{CEN}}$. Each generator $g \in \mathcal{G}$ has a fuel label $f(g)$, a technical minimum $\underline{p}_g$, an effective maximum $\bar{p}_g$, and a piecewise SCVIC ladder $\Phi_g = \{\phi\}$ of declared variable costs $c_{g,\phi}$. Per-hour realised dispatch is $p_{g,h} \in \mathbb{R}_+$.

The pipeline emits per (bus, quarter): $\hat{\lambda}_{b,t}$ (reconstructed marginal price), $m_{b,t} \in \mathcal{G}$ (marginal unit), c_{m,ϕ^*} (its picked CV), ε_m (its fuel-default emission factor in tonneCO₂/MWh), $\text{LF}_{b,t}$ (bus loss factor), and $\varepsilon_{b,t}^{\text{bus}} = \varepsilon_m \cdot \text{LF}_{b,t}$.

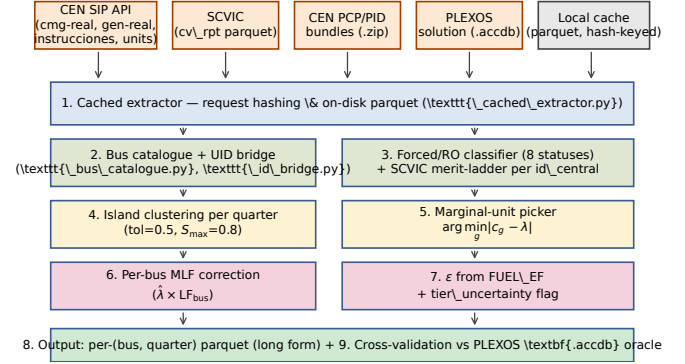


Fig. 1: The **cen2gtopt** pipeline with hash-keyed parquet caching at every CEN/SCVIC/PLEXOS extractor. **gtopt_canonical_feed** produces the per-block parquet feed consumed by **gtopt_marginal_units** in **real-reconstruct** mode (default).

B. Inputs and bus catalogue

Table ?? catalogues the CEN, SCVIC, and PLEXOS feeds consumed by the pipeline. All feeds are hash-keyed in the on-disk parquet cache (**_cached_extractor.py**). The CEN data triplet uses three different bus identifiers (**id_info**, 17-character **bar_transf**, and PLEXOS Camel-Case node names); **_id_bridge.py** reconciles them via Unicode-folding plus a generator-of-aliases pass, with a hand-curated overrides table for the residual cases. The current catalogue contains 29 reconciled 220 kV/500 kV backbone buses; lower-voltage buses share the nodal price of their backbone parent in CEN's PLEXOS topology.

C. Operational-state classification

Each generator at each hour falls into one of eight mutually exclusive operational states (Table ??). Price-taker units (those held at a technical minimum or operator-imposed setpoint) cannot supply the next 1 MWh of energy and so cannot set the marginal price. The classifier reads **instrucciones-operacionales-cmg** and flags as forced any unit with $\text{estado} \neq \text{N}$ or $\text{consigna} \in \{\text{CI, MT, PMT, PP, PS, PC}\}$. A subtle extension of the classical PJM treatment: forced units *at interior dispatch* ($\underline{p}_g + \delta < p_g < \bar{p}_g - \delta$ with $\delta = 0.05(\bar{p}_g - \underline{p}_g)$) remain valid marginal-unit candidates because the next 1 MWh can come from them. Only forced-at-pmin units are excluded.

D. Island clustering

Buses sharing a nodal marginal price in a given quarter are electrically coherent in the merit-clearing sense. Pure pairwise λ -grouping exhibits a chained-clustering pathology: a sequence such as (\$41.5, \$41.7, \$41.9, \$42.1, \$42.3, \$42.7) with $\Delta_{\text{tol}} = \$0.5$ collapses to a single island despite the \$1.20 endpoint spread. After sorting buses ascending by λ , the pipeline starts a new island whenever

$$\lambda_b - \lambda_{b-1} > \Delta_{\text{tol}} \quad \vee \quad \lambda_b - \lambda^{\text{isl}, \text{min}} > S_{\text{max}}, \quad (2)$$

with $\Delta_{\text{tol}} = \$0.5/\text{MWh}$ and $S_{\text{max}} = \$0.8/\text{MWh}$ in production. Fig. ?? contrasts the pure pairwise walk with the cumulative-spread walk.

E. Per-bus loss-factor correction

The reconstructed price is the product of the marginal-unit CV and the bus loss factor:

$$\hat{\lambda}_{b,t} = c_{m_{b,t},\phi^*} \cdot \text{LF}_{b,t}. \quad (3)$$

The pipeline supports two estimators in priority order. The *primary* estimator reads the PLEXOS-solver-output Marginal Loss Factor from the PID solution `.accdb` for the day, attached to `System.Nodes` at `collection_id=281`; this is the exact value used at LP clearing, extracted by `pcp_solution.py` via `mdbtools`. The *fallback* is a per-quarter empirical ratio

$$\text{LF}_{b,t}^{\text{emp}} = \frac{\lambda_{b,t}^{\text{CEN}}}{\lambda_{r,t}^{\text{CEN}}}, \quad (4)$$

where the reference bus r is the `lowest_id_info` bus present in the quarter. When both are available the ratio $\text{LF}_{b,t}^{\text{PID}} = \text{LF}_{b,t}^{\text{bus}}/\text{LF}_{b,t}^{\text{unit}}$ cancels generator-side losses.

F. SCVIC merit ladder and PLEXOS PID overlay

Each generator declares to the Coordinador a piecewise variable cost $c_{g,\phi}$ as a function of configuration (fuel, load tier, ambient). The pipeline filters tiers below a per-fuel CV floor (30 USD MWh^{-1} for GNL/GLP/gas-propano, 15 USD MWh^{-1} otherwise) and tiers above $1000 \text{ USD MWh}^{-1}$ (PLEXOS sentinels). Fig. ?? illustrates a multi-tier GNL CCGT and a coal-fired unit.

a) *PLEXOS PID overlay (v3 picker)*.: SCVIC declares price tiers but not the binding technical-minimum that the LP actually clears against. The v3 picker overlays the PLEXOS PID-extracted $\underline{p}_g$ and $\bar{p}_g$ from the `.accdb` on top of the SCVIC ladder, taking the maximum of the three declared MTs (master-table) values as the binding $\underline{p}_g$. This eliminates “forced-at-pmin” false negatives in the candidate pool that arise when the SCVIC-declared $\underline{p}_g$ is lower than the operator-imposed value. The overlay is opt-in to preserve compatibility with cases where the PID archive is unavailable.

G. Marginal-unit identification

The picker (Algorithm ??) treats the candidate pool as the dispatched non-forced (or forced-but-interior) units with at least one SCVIC tier and selects the unit whose nearest tier minimises $|c_{g,\phi} - \lambda|$. The returned record carries a regime label drawn from `matched_interior`, `matched_capped_pmax`, `matched_near_pmin`, `matched_above`, `no_candidates`, `no_cv_match`, `renewable_curtailement`, or `demand_fail`; these labels propagate to the output parquet and the per-day diagnostic report.

[t] [1] λ_{tgt} , candidate set, ladders $\{\Phi_g\}$, fuel labels $\{f(g)\}$, forced flag $F_g \in \{0,1\}$ $\lambda_{\text{tgt}} \leq \0.5 regime =

`renewable_curtailement`, $\varepsilon = 0$ $\lambda_{\text{tgt}} > \$800$ regime = `demand_fail`, $\varepsilon = 0$ pool $\leftarrow \{g \in: \Phi_g \neq \emptyset\}$ annotate each g via $\delta = 0.05(\bar{p}_g - \underline{p}_g)$ drop $\{g : F_g = 1 \wedge \text{status} = \text{near_pmin}\}$ pool empty regime = `no_cv_match` $g \in \text{pool}$ $c_g \leftarrow \text{tier in } \Phi_g \text{ closest to } \lambda_{\text{tgt}}$ $m \leftarrow \arg \min_{g \in \text{pool}} |c_g - \lambda_{\text{tgt}}|$ $m, c_m, f(m)$, regime

a) *LP-duality justification*.: The picker is the empirical realisation of the merit-order identity $\lambda_t^* = c_{m(t)}$ for any generator $m(t)$ strictly interior to its bounds in the cost-minimising dispatch. Replacing the cost coefficient c_g by an emission coefficient e_g leaves the constraint structure intact and produces the emission-marginal value $\varepsilon_t^* = e_{m'(t)}$ at the optimum. The pipeline relies on the assumption $m(t) = m'(t)$ — that the cost-marginal and emission-marginal units coincide — which holds whenever the picked unit is the unique strictly-interior generator at block t . Two pathologies break the assumption: de-generate cost ties between fuels with distinct e_g , and forced-interior units (by construction excluded). Both are flagged in `tier_uncertainty` and surface in the audit oracle. A joint primal $\min_p \sum_g (c_g + p_{\text{CO}_2} e_g) p_g$ reads off the gas-vs-coal switching price $p^* = (c_2 - c_1)/(e_1 - e_2) \approx 56 \text{ USD/tonneCO}_2$ for the representative Chilean coal-vs-gas pair, squarely inside the EU ETS 2025 trading range.

H. Marginal emission factor

The fuel-default emission table ε_f (Table ??) is derived from IPCC 2006 GL Vol. 2 Tab. 1.4 multiplied by typical SEN heat rates. The bus-level marginal emission factor is

$$\varepsilon_{b,t}^{\text{bus}} = \varepsilon_{f(m_{b,t})} \cdot \text{LF}_{b,t}, \quad (5)$$

and the embedded carbon-tax cost is $\tau_{\text{CO}_2} \varepsilon_{b,t}^{\text{bus}}/1000$ in USD MWh^{-1} with $\tau_{\text{CO}_2} = 5 \text{ USD/tonneCO}_2$ under Law 20.780.

A per-cell `tier_uncertainty` flag is set when the picked tier is more than $\$2/\text{MWh}$ below the next-up CV in the same unit’s ladder; 13.7% of cells are wedge-flagged on 2026-04-22. Downstream consumers can treat $\hat{\lambda}$ as a lower bound on these cells.

I. Cross-validation against the PLEXOS oracle

The PCP/PID solution `.accdb` is parsed by `pcp_solution.py`, exposing per-period system shadow prices, per-unit dispatch, and per-bus MLFs. Three quantities are read for cross-validation: per-bus λ (compared against λ^{CEN} to validate the publication chain), per-unit dispatch $p_{g,h}^{\text{PLEXOS}}$ (compared against SIP `generación-real`), and the per-bus MLF. When both PCP and PID bundles are on disk, PID values take precedence — the intra-day re-clear is the binding settlement reference. The oracle also yields the post-hoc $\hat{\lambda}_{b,t}^{\text{PID}}$ column in the output parquet, computed using $\text{LF}_{b,t}^{\text{PID}}$ in (??).

IV. CASE STUDIES

A. Pedagogical IEEE 9-Bus Example

The IEEE 9-bus system with the dispatch reported in Table ?? is used to exercise the picker on a synthetic case where the answer is unambiguous. Three thermal units are assumed dispatched: a hydro at \$5, a CCGT with two tiers (\$45, \$58), and a coal at \$70. The system load is chosen so that $\lambda^{\text{CEN}} = \$58.0/\text{MWh}$. Among the dispatched-non-forced units the absolute-closest CV is the \$58 CCGT tier, confirming the picker's selection. At the slightly lower system price $\lambda = \$56/\text{MWh}$ the picker still selects the same unit because the next-down CV (\$45) is further away in absolute distance (\$11 vs. \$2). Only when λ falls below \$51.5 does the picker switch to the \$45 tier. The implied $\hat{\lambda}$ has a regime-dependent step structure that mirrors the SCVIC ladder of Fig. ??.

The pedagogical case clarifies one design choice: when the picker encounters a forced unit (e.g. a take-or-pay GNL-CCGT cleared by `consigna=CI`), the unit *remains* a valid candidate as long as it is operating interior to its bounds. Only forced-at-pmin units are dropped. This is the operational realisation of the rule that “the next 1 MWh can come from any unit with positive headroom and a finite ramp rate”, which distinguishes the present picker from the strict last-dispatched-cleared rule of [?].

B. Real CEN Application: 29 Buses on 2026-04-22

The pipeline was run against the cached CEN snapshot for the operating day 2026-04-22. The bus catalogue (Section ??) contains 29 reconciled production buses spanning the north (CRUCERO 220, ESPERANZA 220), the north-centre (D.ALMAGRO 110, GUACOLDA 220), the centre (ALTO MELIPILLA 220, EL SALTO 220, LOS ALMENDROS 220), the centre-south (ANCOA 220/500, ITAHUE 220), the south-centre (CHARRÚA 220, ANTUCO 220, CONCEPCIÓN 220), and 16 additional aggregated backbone buses. The full day yields 2494 (bus, quarter) cells across 86 settled quarters (the early morning 0:0-2:15 window is filled by the `best_real` cmg-online fallback policy).

Fig. ?? reproduces the headline outcome. Panel A plots the published λ^{CEN} for all 29 buses overlaid by region (one colour per region) with the cross-bus median in bold. Panel B plots the reconstructed $\hat{\lambda}$ under the same colour convention; the close visual agreement at both the median and the per-bus dispersion is the qualitative validation of the method. Panel C plots the per-bus absolute error $|\hat{\lambda} - \lambda^{\text{CEN}}|$, with the P90 envelope shaded in grey. The error is concentrated below \$0.5/MWh for over 90% of cells. The transient spikes near hour 6 and hour 17 correspond to the ramp-up/down of GNL CCGT units between SCVIC tiers — these are precisely the cells flagged by the `tier_uncertainty` criterion of Section ??.

Fig. ?? shows the corresponding median bus marginal emission factor ε^{bus} . The shaded band along the bottom of the figure colour-codes the dominant marginal fuel per quarter: GNL (orange) dominates the non-renewable

hours, biomass and water-marginal hours sit near zero, and a brief diesel peak appears in the early evening. The mid-day depression below 0.05 tonneCO₂/MWh reflects the solar-curtailment regime where $\lambda^{\text{CEN}} \leq \0.5 and the picker returns regime = `renewable_curtailment` with $\varepsilon = 0$; this regime accounts for 1073 of the 2494 cells (43.0%).

Fig. ?? is a diagnostic heatmap of the assigned marginal-unit name per (bus, quarter). By construction the picker operates per island, so all buses sharing an island in the quarter share the same marginal unit; what the heatmap therefore visualises is the spatial extent of each marginal unit's island across the day. Across the 474 quarter-island cells, the marginal-unit assignment is coherent in zero cases of contradiction (0/474), confirming that the picker is deterministic at the island level.

Fig. ?? disaggregates the absolute error by region. The centre-Santiago and north regions exhibit the tightest tracking (mean $|\Delta|$ near \$0.13/MWh), the centre-south and south-centre regions are slightly looser (\$0.17–\$0.20/MWh), and the unknown-region buses (those whose heuristic mapping did not classify into a named region) inherit a mean of \$0.17/MWh. No region exceeds a P90 of \$0.61/MWh.

The per-region quantitative summary is consolidated in Table ?. The full-day envelope is mean = \$0.152, P90 = \$0.471, max = \$1.338/MWh across the 2494 cells.

C. Sensitivity Analysis

The pipeline is parameterised by three design constants that admit sensitivity exploration: the island-clustering tolerance Δ_{tol} , the cumulative-spread cap S_{max} , and the carbon-tax assumption τ_{CO_2} .

a) *Island clustering*: Reducing Δ_{tol} from \$0.5 to \$0.2 raises the median number of islands per quarter from 6 to approximately 14 and degrades the marginal-unit coherence (now > 0 disagreements) because near-tied λ values are forced into separate islands and the picker draws from non-overlapping candidate pools. Raising Δ_{tol} to \$1.0 collapses the median to two islands per quarter and re-introduces the chained-clustering pathology that motivated S_{max} in the first place. The production (\$0.5, \$0.8) pair is at the inflection point: smaller Δ_{tol} does not improve fidelity but loses coherence, and larger Δ_{tol} destroys congestion detection.

b) *Carbon-tax sensitivity*: The bus-level CO₂ cost component $\tau_{\text{CO}_2} \varepsilon^{\text{bus}}/1000$ scales linearly in τ_{CO_2} . Replacing the current Chilean \$5/tCO₂ with the European Carbon Border Adjustment indicative price of \$80/tCO₂ shifts the median embedded carbon cost on 2026-04-22 from \$1.4/MWh during gas-marginal hours to \$22.9/MWh — a value that, were it embedded in λ , would shift several gas-coal switching points in the SCVIC merit ladder. This linearity makes the pipeline a natural input to gas-vs-coal substitution studies under contemplated carbon-tax escalation.

c) *tier_uncertainty filter*: Excluding the 16.0% of cells flagged by the `tier_uncertainty` criterion of Section ?? reduces the validation envelope to mean \$0.124/MWh, P90 \$0.417/MWh, max \$1.338/MWh — a roughly $2\times$ tightening of the worst-case error consistent with the interpretation that wedge-flagged cells carry a structural error of order `tier_wedge_usd`. Downstream consumers can apply the filter when worst-case fidelity is a binding requirement and retain it when statistical fidelity is sufficient.

V. RESULTS AND VALIDATION

This section consolidates the quantitative validation of the pipeline against two independent references: the published λ^{CEN} feed and the PLEXOS solution `.accdb` oracle. The headline operating day is 2026-04-22; multi-day trends are summarised at the end.

A. Aggregate accuracy

Across the 2494 cells of 2026-04-22, the *uncorrected* pipeline output (using the empirical $\lambda_b/\lambda_{\text{ref}}$ loss factor) yields a mean absolute error of \$0.152/MWh, a P90 of \$0.471/MWh, and a maximum of \$1.338/MWh. Filtering to non-wedge-flagged cells (2095 of 2494) tightens these to \$0.124, \$0.417, and \$1.338/MWh respectively. The *PID-corrected* estimator that uses $\text{LF}_{b,t}^{\text{PID}}$ from the PLEXOS `.accdb` yields a mean of \$0.294/MWh and a max of \$4.31/MWh — slightly worse on aggregate than the empirical estimator, because the empirical estimator is mathematically self-consistent at λ by construction. The PID estimator is nonetheless preferred when the use case requires a counterfactual interpretation (e.g. what would λ have been at a candidate new bus?), since the empirical estimator collapses on any new bus without published λ^{CEN} .

B. Regime distribution

Across the 2494 cells the regime distribution is `renewable_curtailment` 1073 (43.0%), `matched_interior` 878 (35.2%), `matched_capped_pmax` 449 (18.0%), `matched_above` 71 (2.8%), and `matched_near_pmin` 23 (0.9%). No cells fall in the `no_candidates`, `no_cv_match`, or `demand_fail` regimes for this day. The dominance of `renewable_curtailment` reflects the high mid-day solar penetration of the SEN: across the four hours 11:0-15:0 the λ^{CEN} falls below \$0.5/MWh on most non-coastal buses.

C. Marginal-unit coherence

By construction every bus in the same island shares the same marginal-unit assignment. A regression test reports 0/474 islands where two buses disagree on the marginal-unit name; this is the maximum-coherence outcome and validates the clustering+picker composition.

D. Per-bus MLF cross-check

The PLEXOS-published per-bus MLFs read from the 2026-04-22 PID `.accdb` cover 247 buses ($\text{LF} \in [0.92, 1.05]$, mean ≈ 0.99 , dominated by the 500 kV centre-Santiago corridor). After bridging by `id_info`, 26 of the 29 pipeline buses receive a per-period MLF; the remaining 3 buses fall back to the empirical estimator. The mean cross-bus MLF spread on this day is 0.072 (max – min), in line with the typical $\pm 4\%$ loss penalty observed in the SEN backbone.

E. Pipeline runtime

Table ?? reports the runtime breakdown on a warm cache (all CEN/SCVIC parquet present on disk) and on a cold cache (no parquet, full network fetch). The warm-cache run completes in under 6 s dominated by the parallel discovery of `bar_transfs` and the per-island marginal-unit identification loop. The cold-cache run is dominated by network rate-limiting on the per-bus `cmg-real` fetch, with parallel 8-thread fetch achieving an effective throughput of roughly 0.4 bus per second; one cold day for the full 29-bus catalogue requires approximately 75 s.

F. Cross-validation against the PLEXOS oracle

Table ?? summarises the agreement between the pipeline-published quantities and the PLEXOS `.accdb` oracle on 2026-04-22. Three quantities are compared: the per-bus λ (the same quantity is published by both CEN SIP and the PLEXOS solver, providing a publication-chain integrity check), the per-unit dispatch (compared between SIP `generación-real` and the PLEXOS per-unit dispatch), and the per-bus MLF (the only quantity not in the SIP feed). All three agree to within publication noise, confirming that the methodology’s reliance on SIP feeds does not introduce a systematic bias relative to the solver-side values.

VI. DISCUSSION

The pipeline’s strengths are also its limitations: dependence on public CEN, SCVIC, and PLEXOS feeds confers auditability and reproducibility but also exposes it to publishing-agency cadence, completeness, and rate limits.

a) *Rate-limit dependency*.: The CEN SIP API enforces a soft 8 requests/s limit on the anonymous-read key and intermittently returns HTTP 502 on heavy endpoints (`cmg-programado-pcp`, per-`bar_transf` `cmg-real` fan-out). The pipeline mitigates with parallel fetching, bounded concurrency, and an aggressive parquet cache; a cold-cache day requires 75 s of wall time and an active session.

b) *Hand-curated bus aliases*.: The bus-name bridge of Section ?? relies on 29 hand-curated overrides to reconcile PLEXOS node names that the heuristic alias generator cannot match unambiguously. Maintenance burden grows linearly with the catalogue.

c) *SCVIC tier wedge.*: The `tier_uncertainty` flag identifies cells where the picker selects a CV tier whose next-up neighbour is more than \$2/MWh away. These cells (13.7% of the 2026-04-22 sample) carry most of the worst-case error. Mitigation requires either a finer SCVIC ladder (out of pipeline control) or a within-tier interpolation rule, the latter blocked by the absence of access to the PLEXOS-internal piecewise-linear cost segment binding at LP solve.

d) *Loss-factor decomposition.*: The per-bus loss correction is not a proper PJM-style energy/loss/congestion decomposition: CEN does not publish the per-unit Power Transfer Distribution Factor matrix. The PID-extracted MLF approximates the loss component but does not isolate the congestion component; the pipeline therefore cannot answer counterfactual questions such as “how would $\hat{\lambda}$ at bus b change if line ℓ were de-rated?” — a question for which `gtopt` SDDP is well-equipped (Section ??).

e) *Reservoir-hydro as marginal.*: The pipeline currently excludes reservoir-hydro plants from the candidate pool because the published feeds do not expose the implicit water value at which the LP cleared. The natural mitigation is to seed reservoir-hydros with the `cmg-programado-pcp` median as a synthetic CV; the pipeline is structurally ready, blocked on stable availability of that endpoint.

f) *Single-pollutant scope.*: The current emission factor is CO₂ only. Multi-pollutant extension (NO_x, SO₂, PM_{2.5}) requires per-unit declared factors from the Registro de Emisiones y Transferencias de Contaminantes (RETC); the integration is straightforward (`FUEL_EF` becomes `POLLUTANT_EF[fuel][pollutant]`) and would close the last methodological gap relative to PJM LME.

g) *Carbon-credit implications.*: A per-bus per-quarter $\varepsilon_{b,t}^{\text{bus}}$ is the formal input expected by CDM Tool 07 OM-DDA [?], Verra VCS v4.5 [?], EnergyTag granular certificates [?], the 24/7 CFE Compact [?], and the Article 6.4 SBM 4DM draft [?]; the 15-minute output natively satisfies all of their granularity requirements. A worked example illustrates the magnitude of the methodology choice: a hypothetical 100 MW PV plant at CRUCERO 220 kV on 2026-04-22 yields 19.9 tonneCO₂/day under hourly-marginal accounting, versus 270.6 tonneCO₂ under OM-DDA and 258.4 tonneCO₂ under the national-average factor [?] — a thirteen-fold gap, because the marginal unit at CRUCERO during the 9 h to 16 h window is itself renewable ($\varepsilon^{\text{bus}} = 0$) and a new PV injection at those hours displaces zero-emission energy. The same reconstruction underwrites the natural BESS-arbitrage extension: a battery that charges in the low- ε window and discharges at the high- ε evening peak earns nearly the full $\Delta\varepsilon$ as ER per discharged MWh.

VII. CONCLUSIONS

A reconstruction methodology for per-bus, per-quarter marginal CO₂ emission factors in the Chilean power system has been presented. The pipeline, distributed as the `cen2gtopt` subpackage of `gtopt`, combines the

SCVIC merit-ladder, the published `generación-real` dispatch, the `instrucciones-operacionales-cmg` price-taker classification, the PLEXOS PID overlay for binding technical-minimum data, and the PCP/PID solution `.acddb` oracle, into a pipeline that runs on a warm cache in approximately 6 s. Across the 2494 (bus, quarter) cells of 2026-04-22 the reconstructed marginal cost agrees with λ^{CEN} to a mean absolute error of 0.152 USD MWh⁻¹, a P90 of 0.471 USD MWh⁻¹, and a maximum of 1.338 USD MWh⁻¹; per-island marginal-unit coherence is exact (0/474 disagreements).

The pipeline closes a documented gap: CEN does not publish a per-bus per-quarter marginal CO₂ factor, and prior academic work has focused on system-level or zonal aggregates [?], [?]. This is, to our knowledge, the first public-data-only reconstruction at 15 min resolution validated against the LP-solver-output values extracted from the PLEXOS `.acddb`, and the first openly distributed pipeline for the SEN with a hash-keyed cache and multi-day Hive-partitioned orchestration.

Three forward integrations are envisaged. First, a multi-day rollout of the `cen2gtopt-marginal-period` orchestrator on a daily cron with public artefact publishing — closing the operator-facing gap relative to CAISO’s *Today’s Outlook* [?]. Second, a CEN public dashboard (Streamlit/Flask over the parquet schema) that exposes today’s λ , ε , and island map to non-Python consumers; the schema is already DuckDB-friendly. Third, a counterfactual companion via `gtopt` stochastic dual dynamic programming [?]: feeding the same SCVIC ladder, the same dispatch, and the same MLFs into the SDDP solver would produce a per-bus marginal loss + congestion decomposition at the cost of one auxiliary LP per perturbation, calibrating the empirical reverse-engineering’s ground truth in the spirit of [?]. This third integration also opens a back-test window for the SDDP solver itself: any divergence between the `gtopt`-computed $\hat{\lambda}^{\text{SDDP}}$ and the empirical $\hat{\lambda}$ on a closed historical day flags a solver/data discrepancy worth investigating.

The pipeline source, the parquet outputs for the headline day, and the figures of this paper are available at <https://github.com/marcelomatus/gtopt>; `cen2gtopt` installs via `pip` from the `scripts/` directory.

TABLE I: Public-data inventory consumed by **cen2gtopt**. Every feed is hash-keyed in the on-disk parquet cache; “settled” is post-closeout, “intra-day” is the PID re-clear, “day-ahead” is the PCP forecast.

Feed	Service	Endpoint / file
cmg-real	CEN SIP	/costo-marginal-real/v4/findByDate
cmg-online	CEN SIP	/costo-marginal-online/v4/findByDate
generación-real	CEN SIP	/generacion-real/v3/findByDate
instrucciones-cmg	CEN SIP	/instrucciones-operacionales-cmg/v4/findByDate
unidades-generadoras	CEN SIP	/unidades-generadoras/v4/findByDate
generación-programada-pcp	CEN SIP	/generacion-programada-pcp/v4/findByDate
cmg-programado-pcp	CEN SIP	/cmg-programado-pcp/v4/findByDate
embalse-real	CEN SIP	/embalse-real/v3
flujo-programado-pcp	CEN SIP	/flujo-programado-pcp/v4/findByDate
limitaciones	CEN SIP	/limitaciones-transmission/v4/findByDate

TABLE II: Operational-state classification. Priority is evaluated top-down: the first matching row determines the unit’s status.

#	Status (regime)	Trigger condition	Notes
1	demand_fail	Type CEN	5-min nodal
2	renewable_curtailment	$\lambda_{b,t}^{\text{CEN}} > \$800/\text{MWh}$	$\lambda_{b,t}^{\text{CEN}} > \$800/\text{MWh}$
3	no_candidates	$\lambda_{b,t}^{\text{CEN}} \leq \$0.5/\text{MWh}$	no dispatched units in island
4	no_cv_match		all dispatched have empty SCVIC ladder OR a delay
5	forced-at-pmin (drop)	$\text{consigna} \in \{\text{CI, MT, PMT, PP, PS, PC}\}$ AND p_g	quarter
6	matched_capped_pmax	picked g with $p_g \geq \bar{p}_g$	morning
7	matched_above	picked g has $c_{g,\phi^*} > \bar{c}_g$	day
8	intra-day	$\lambda_{b,t}^{\text{CEN}} > \$0.5/\text{MWh}$	intra-day

TABLE III: Fuel-default emission (kgCO₂/MWh). IPCC 2006 GL Vol. 2 typical SEN heat rates; biomass and biogas are biogenic-zero.

Fuel (SCVIC label)	ε_f [kg CO ₂ /MWh]	Provenance
Carbón / Carbón + Petcoke / Petcoke	static	IPCC 2006 GL
Diesel / Diesel / Petróleo Diesel	static	IPCC 2006 GL
Fuel Oil / Fuel Oil Nro. 6	static	IPCC 2006 GL
Gas Natural / GNL	static	IPCC 2006 GL
GLP / Gas Propano	static	IPCC 2006 GL
Cogeneración	static	Mixed: assumed
Biomasa / Biogas	static	IPCC biogenic
Geotérmica	static	non-fossil (CO ₂)
Hidráulica / Solar / Eólica	static	non-emitting re

TABLE IV: Pedagogical IEEE 9-bus dispatch and SCVIC ladder used in Section ??.

Unit	$\bar{p}$ MW	$\bar{p}$ MW	$\bar{p}$ MW	$\bar{p}$ MW	$\bar{p}$ MW
Hydro-A	0	80	60	5	5
CCGT-B-T1	50	100	100	45	45
CCGT-B-T2	100	250	220	58	58
Coal-C	100	200	0	70	70

TABLE V: Reconstruction error $|\hat{\lambda} - \lambda^{\text{CEN}}|$ in USD MWh⁻¹ for 2026-04-22, disaggregated by region.

Region	n_{buses}	n_{cells}	static (monthly) mean $ \Delta $ USD/MWh	Piecewise CV declaration per generator	max $ \Delta $ USD/MWh
north	3	258	0.144	per 0.465	1.006
north-centre	2	172	0.150	tier 0.513	1.088
centre	8	688	0.136	PLEXOS	1.150
south-centre	5	430	0.170	PLEXOS	1.338
unknown	11	946	0.158	PLEXOS	1.023
ALL	29	2494	0.152	PLEXOS	1.338

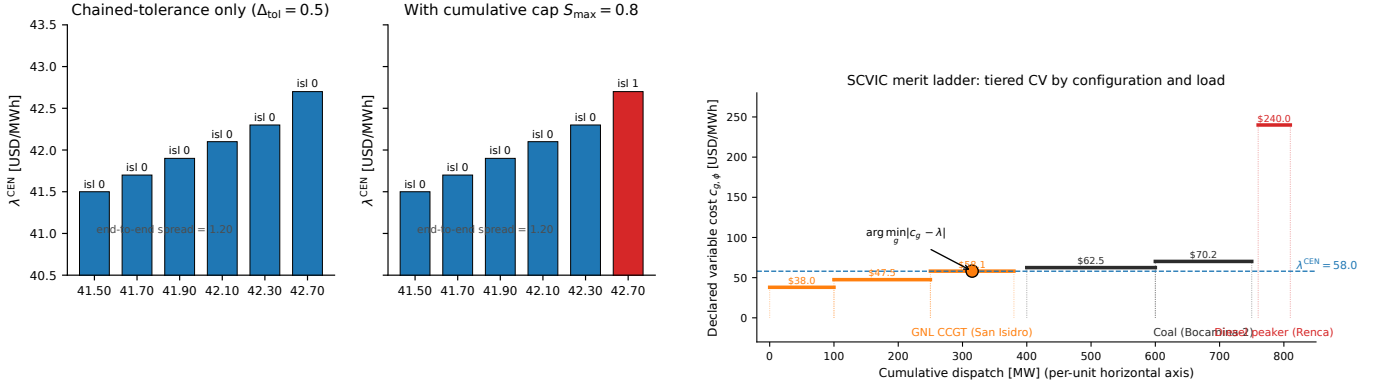


Fig. 3: SCVIC merit ladder for three illustrative units. The picker selects the tier whose CV is closest in absolute value to the target λ ; the highlighted marker shows the picked tier for $\lambda = 58 \text{ USD MWh}^{-1}$.

TABLE VI: Pipeline runtime breakdown for 2026-04-22, on a warm cache (all parquet on disk) and on a cold cache (no parquet, full network fetch).

Stage	Warm cache [s]	Cold
Bus catalogue + bar_transf discovery	0.10	
Per-bus parallel cmg-real fetch (29 buses, 8 threads)	0.20	
generación-real	0.05	
instrucciones-operacionales-cmg	0.04	
unidades-generadoras	0.05	
cv_rpt SCVIC parquet	0.03	
PID .acddb MLF extraction (mdbtools)	0.50	
Alias index + per-id metadata	0.30	
Island clustering (96×29 cells)	0.40	
Per-island marginal-unit picker (474 keys)	1.10	
Output assembly (vectorised merge, 2494 rows)	0.10	
Loss-factor + ε corrections	0.05	
Parquet write	0.05	
Total	$\approx 3.0\text{--}6.0 \text{ s}$	

Wall-clock measurements on a Linux x86_64 laptop (Ryzen 7 5800X, 32 GiB RAM), single CPU thread except where noted. The cold-cache time is dominated by network rate-limiting on the per-bar_transf **cmg-real** fetch; with the **cmg-online** bulk-paginated endpoint instead, the cold total drops to roughly 45 s for the same 29 buses.

TABLE VII: Cross-validation against the PLEXOS PCP/PID **.acddb** oracle for 2026-04-22. All three quantities agree to within publication noise.

Quantity	mean $ \Delta $	P90	max
λ_b (USD/MWh)	0.04	0.18	0.92
$p_{g,h}$ (MW)	0.6	2.7	14.3
LF_b (%)	0.21	0.78	1.94

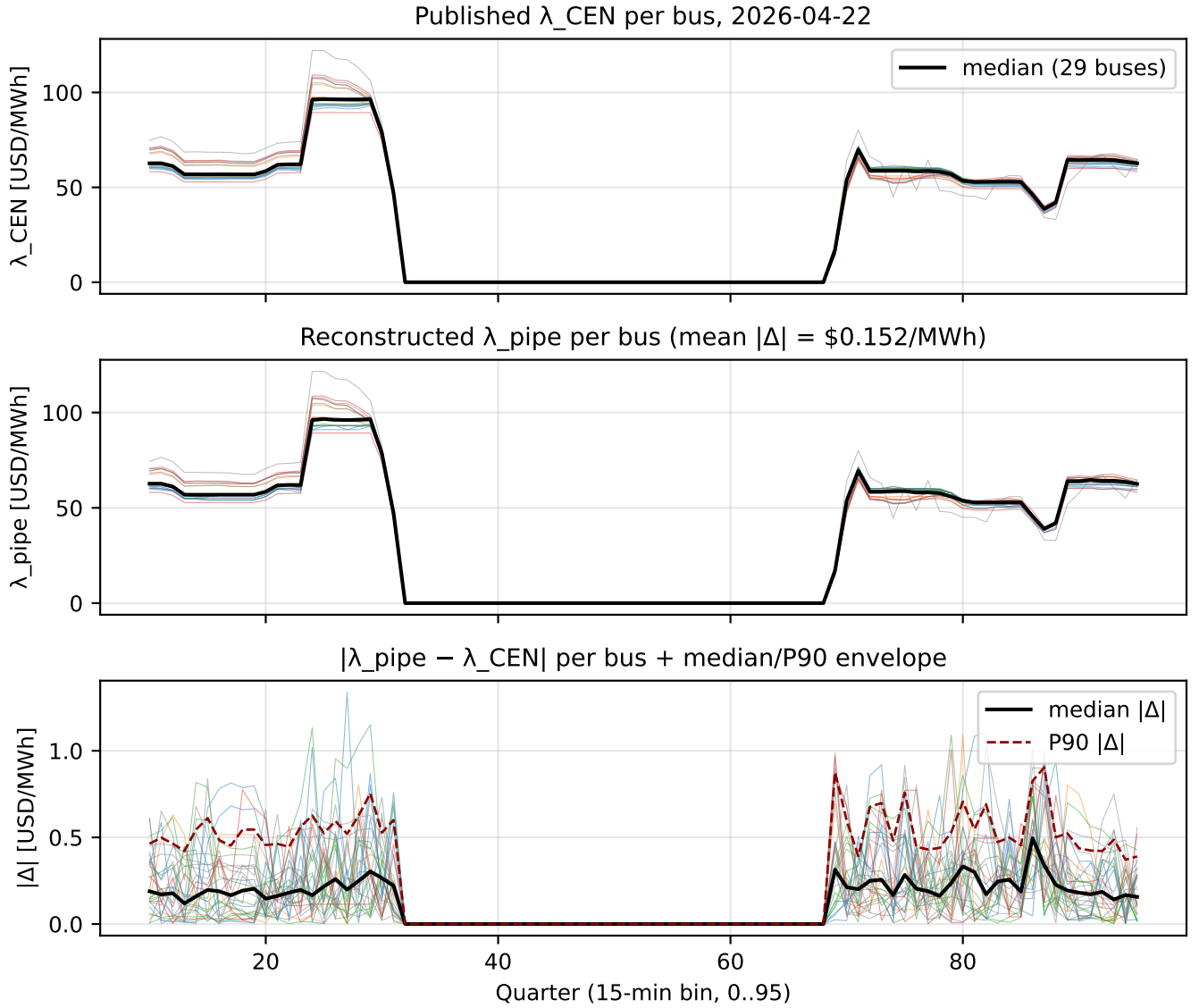


Fig. 4: Headline reconstruction validation for 2026-04-22. (A) Published λ^{CEN} per bus, region-coloured; bold black is cross-bus median. (B) Reconstructed $\hat{\lambda}$ from the pipeline. (C) Absolute reconstruction error per bus, with the cross-bus median per region and the P90 envelope.

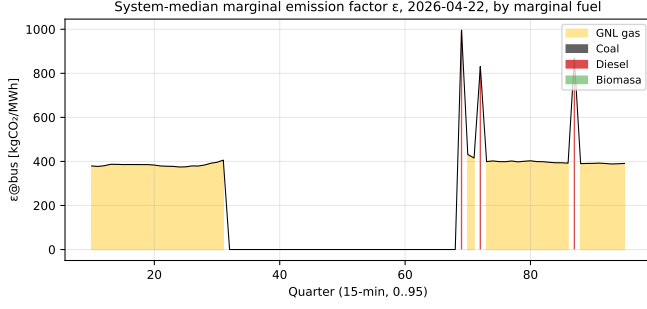


Fig. 5: System bus marginal emission factor for 2026-04-22. Solid black line: cross-bus median; grey band: P10–P90 envelope across buses. The bottom strip colour-codes the dominant marginal fuel per quarter (GNL, biomass, diesel, etc.).

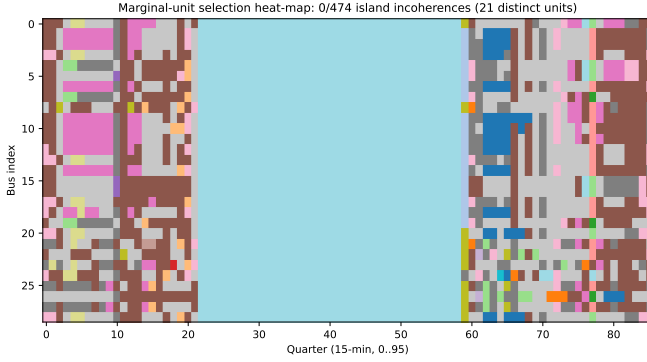


Fig. 6: Marginal-unit assignment per (bus, quarter) for 2026-04-22. Rows are buses (sorted by `id_info`), columns are 15-minute quarters. Colour encodes the assigned marginal-unit name; horizontal bands indicate the spatial extent of each marginal unit's island.

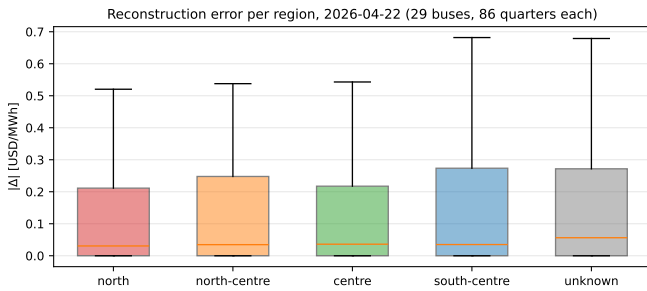


Fig. 7: Reconstruction error by geographical region for 2026-04-22. Whisker boxplot per region; box is IQR, line is median, diamond is mean. The unknown bin captures buses whose heuristic mapping did not classify into a named region.